ActInf ModelStream 012.1 ~ Fisher, Whyte, and Hohwy: An Active Inference

ModelStream | Fisher, Whyte, and Hohwy: An Active Inference Model of the Optimism Bias | 1:05:41

all right hello and welcome it is June
27th Z UTC 2024 and we are in active
inference model stream number
12.1 with Beth fiser and yaka howi so we
will have a presentation and a
discussion on an active inference model
of the optimism bias so thank you both
for joining and Beth to you for the
presentation great thanks Daniel um yep
so I'm Beth and I'm going to talk about
work we've just put out as a pre-print
and active inference model of the
optimism bias and this is work we've
done along with Chris White who I think
a lot of you will already know or read
some of his work and yeah we're here in
Australia at the Mones Center for
Consciousness and contemplative studies
and this is part of my PhD project so
I'm doing my PhD with
yako so what I'll do is I'll first of
all give a bit of of an overview of what
the optimism bias is and why this is
interesting to study why I'm interested
in it I'm then going to talk a bit about
um the field of computational Psychiatry
so I'm using active inference not so
much in a way to like to understand the
nitty R of active inference but more as
an application as a tool in
computational Psychiatry so I thought it
would be helpful to explain how we can
use these models in in these kinds of
settings and and the purpose of why I'm
using these models then I'll go through

the simulations uh that we have in the
paper and then just some
conclusions so the optimism bias is a
cognitive bias where our expectations
are better than reality
and what's interesting about the
optimism bias is that you might think in
order to make the best decisions in life
and have the best life outcomes you
should represent information as it is in
reality and not have expectations that
are better than reality but the optimism
bias shows us that if we think things
are better than they are and
overestimate the likelihood of good
things happening we have all these
better life
outcomes so people who are optimistic
have a lower risk of all CA ality and
reduce risk of cardiovascular
disease they have decreased fatigue and
chronic illness and they engage in more
self-care
behaviors there's even this really cool
study that showed that people who are
optimistic had improv cognitive
functioning following a traumatic brain
injury um optimists they just earn more
money they have better immune systems
and they just have less anxiety and
worry um
overall so why would it be that being
optimistic and thinking things are
better than they are leads to all the
these positive
things so to explain that I'm going to
go through an
example so imagine here you're someone

who holds the reality beliefs so in
reality it's true that 50% of marriages
end in divorce if you work out you
reduce your risk of heart disease by
14% And if you eat well you reduce your
risk of cancer by 10%
but then imagine that you're someone
who's optimistic so if you're optimistic
you might think no only 10% of marriages
and in divorce if I uh work out I'll
reduce my risk of heart disease by 40%
and if I eat well I'll reduce my risk of
cancer by
30% so if we hold these optimistic
beliefs we're going to be way more
likely to do these things so we'll be
more likely to eat well we're going to
be more likely to work out we're going
to be more likely to go out there with
the hope of finding a
partner and although not everything will
work out for you because the reality
statistics are there on the left by
going out and engaging in more
activities and things in life we we
increase our chances of some things
working up it's like the idea of
minimizing the risk of missed
opportunity if I do go out there and
work out and go to parties and eat well
and all these things I'm going to have
more likely chance of some of them
working out than if I just stay at home
and don't do anything so being
optimistic increases our engagement in
the world and this overall is a positive
thing that can lead to all these health
benefits so if optimism is really good

for all these things what is the
connection between optimism and mental
health so people with an anxiety and
depression have been shown to lack
optimism they can either have
expectations that worse than reality so
this can be pessimism or they can have
expectations that match reality so this
is sometimes called depressive
realism which as we've kind of gone over
that this isn't a good thing
either so what I'm interested in is well
could we increase optimism as an
intervention for depression and anxiety
so that's what what I what I what I want
to
do so to explain how I'm going to do
this and and how active inference fits
in into this goal uh I'm just going to
explain a bit about computational
Psychiatry and and what we're kind of
aiming to do in in that
space and I'll go through the steps of
of what you want to do for your aim so
the first thing which I guess in in most
areas in science you have to come up
with a a conceptual
idea so to do this for this project I
read a lot about active inference I read
a lot about optimism and try to think
how active inference could best model
optimism what predictions we could maybe
make from these models and and how we
can model that so you want to spend a
lot of time on this first step coming up
with a good conceptual idea and I'm sure
uh a lot of you who have done the active
inference work you know that there

there's a lot to that that step that's
an important
step but once you come up with your idea
uh you can run these simulations so
that's what we've done in this paper so
thought a lot about optimism how it fits
in active inference and then ran some
things uh on my computer to see okay
well can I simulate what I want to see
am I seeing
optimism come out in active inference in
the way that I want it to and then the
next step after that the idea is that
after you've run your simulations you
kind of have a good base to then fit the
data so what's nice about running the
simulations is you kind of have more
background knowledge before you go out
and just start start fitting and
collecting data so you can design your
experiments to to fit what you found in
in the
simulations um my image here is of these
rats because we have other work where
we've done um active inference data
fitting to to a r model so that's what's
going on
there and then the point of all of this
is that hopefully in the end we can come
up with a model that will have an
application in the clinic so after we've
done our simulation and our data fitting
well the idea to use the optimism as the
example we'll have some model that then
clinicians can use to help measure and
understand how optimism is um working
for a patient and then they can use that
model to to design treatments for that

person so that's the
whole goal and aim of what I'm doing and
that's why I'm doing what I want to do
and that's why and how I'm using active
inference so in terms of the optimism
plan this paper is talking about the
simulation step and then hopefully from
there we can do some data fitting for
this
work so then there are just some other
things I wanted to say before we start
talking about the models about when
we're thinking about models to use in
the clinic there's some
considerations um that are important
that might be different to
considerations if you're using active
inference and modeling in other set
but if we want to um use them in
computational Psychiatry we actually
want simple models because if the end
goal is that a psychiatrist or
psychologist or or or doctor uses them
we don't want anything super complicated
because um that won't be practical that
won't be that won't be used so one of
the things when I'm designing my models
is I'm trying to to use models that that
are simple that can give us more
information about it ass sort
um but is still can still be used and
then the idea is that this can help with
Precision Psychiatry so at the moment um
you can go to the doctor you do a self
report for depression you get diagnosed
with depression but there isn't really
much understanding of the mechanisms
that are involved of of of what caused

the depression or different things that
may have caused it so hopefully the
these models that we design can um we
can understand okay well maybe that
person is experiencing these uh
depressive symptoms because they have
low optimism or maybe they don't have
low optimism maybe there's something
else that that's going on um that we can
understand through the use of these
models so I hope that kind of sets the
scene of of this work and um and and
hopefully that made
sense so when I did all the reading of
the literature and on active inference
and optimism we this is our conceptual
idea of an active inference model of the
optimism bias that is the Precision over
the positive outcomes for the
agent um and that might not sound that
groundbreaking given the definitions of
optimism I've
given uh but we'll go through the models
to see how we can fit this in into the
framework so what we did is that we
formalize optimism as a state factor in
the generative model that then could be
applied to different simulations and
tasks so we have this um I'm not sure if
you can see the mouse very easily but we
have this oh we can um optimism State
Factor here and we have an optimistic
State and a pessimistic State and the
idea is that you can take this state
factor in different situations and
different task settings and we wanted to
do something like this because optimism
is this you know this higher higher

level belief that affects different
areas of life it's it's not just uh
constrained to one one setting so we
wanted to think okay well how can we use
something that can be General and
applied to different different tasks and
can we um use this in the
simulations and the optimism State
Factor controls or it's conditions the
Precision of the updating for the
positive outcomes and we'll go through
the examples and hopefully that will
start making more sense how that works
so when I go through the rest of the
talk I'll refer to optimism bias as
levels and I just wanted to explain what
I mean when I talk about that so we
refer to these different optimism levels
depending on the optimism State Factor
so here we would say this is optimism
level1 um because there it's 0.1
optimism 09 pessimism you will have 0
five optimism and then this would be
fully a level one optimism
sorry um so when I go through the rest
of the presentation this is what when
I'm referring to levels this this is
what I
mean so we ran well we three different
simulations but we we refer to them as
experiments in this
paper so the first one was we simulated
the loss of the optimism
bias so we wanted so one of the things
um that's interesting about
optimism is it Sean across species so we
see it um in Bay in horses all animals
have

optimism and it's this this is it's an
Adaptive adaptive bias and it suggests
that the optimism bias is something that
we're born with it's something it's not
like other biases that that we learn
later on it's something that is is a
part of us and there's a lot of research
showing that optimism um
over if depending on your environment in
childhood you can have lower levels of
optimism so factors that you're exposed
to early on can influence the level of
optimism that you have and we wanted to
kind of do this first experiment with
this developmental stage to simulate how
optimism might be lost or maintained
during development so that's the idea of
the first experiment
the second experiment uh we applied our
optimism by a state factor that was lost
or or maintained in experiment one to a
belief updating task that's used in the
literature on optimism so this is used
in human participants uh to study
optimism and we wanted to do this
because hopefully this can ground our
model in actually empirical literature
and then if we're able to replicate
results that is found in the empirical
literature and optimism it kind of gives
our
model we we kind of have more hope that
this is a model that we can then go out
and use so that was the idea of that
step and then the third step was we ran
a two Arm Bandit task um but it's it's a
modified two Arm Bandit task where we
wanted to see if we could measure

optimistic
action
so um because optimists engage in the
world more we wanted to see if we can
develop a task that can measure that
kind of Engagement so I'll go through
the details of that a bit when we get to
that
step so the first uh simulation I'll go
over is the simulation of the loss of
the optimism
bias so as I've already kind of
mentioned there are environmental
factors that affect optimism level so
this can be um you know things that
happen to you in childhood if you're
exposed to trauma
um your socioeconomic status these kind
of things so people who usually have
gone through more adversity during
childhood have low
optimism and another thing is that
people who are optimistic have improve
cardiovascular responses to stress so
this means that they have a healthier
arousal outcome so if they get really
stressed out or exposed to something
they they don't have this the super high
um AR which which as we know is isn't
very good for
you and then on the other hand as
probably most of you know anxiety is
related to high arousal so if you're
anxious um you can have just higher
arousal overall and also when there's
there's nothing there to really warrant
that that
arousal so what we wanted to do was to

simulate this loss of optimism bias over
time by exposing the agents to higher
and lower rousel and positive and
negative violence
events and um the idea behind that is
that obviously we know that there's uh
higher arousal you can have high arousal
from positive and negative things and
you can have low arousal from positive
and negative things and a healthy agent
would be able to have all of those
equally and and and that's that's a good
thing whereas some someone who is
anxious or they can sometimes experience
this higher arousal um as a negative
thing if even when it's not so this is
what we wanted to see if we expose the
agents to these things during
development does this change their
optimism and we also looked at how it
looked uh affected their arousal
mappings but we'll go through that in
the
figure so for this experiment so this is
just our little image of the experiment
so we have the agent who's exposed to
all of different combinations of these
events so positive high and low negative
high and
low and for this experiment we have a
hierarchical active inference model so
this means that we have two
levels so the first level of this model
just set up the veilance arousal States
so we just fed the model different um
veilance and arousal States and there
was no learning there was no no actions
it was just just feeding those

observations that the that the agent had
so this is our first
level and then on our second level this
is the agents optimism so this is
happening at a slower time time course
they're getting these uh veillance and
arousal observations um over these these
weeks that that's how we operationalize
it and this is changing affecting their
their optimism so in the hierarchical
models um the the states of of level one
they give the observations for level two
so whatever state the agent was in in
this level one it gives them their their
observations
there so on this second level the agent
was learning two things so they were
learning their their optimism which is
um our state Factor here
and they were also learning their
arousal likelihood Matrix so over time
um if they're in an optimistic or
pessimistic State the mapping between
high and low
arousal so they're the two things that
that we were looking
at so our first result so this is what
the optimism looked at after the
simulation of development and we we ran
it for 200 a
so this is optimism this is optimism
State Factor remember we talk about
those levels and then the proportion of
negatively valanced events they
had so what we found is that agents that
were more optimistic after development
were exposed to less negative vean
events which is what we would hope to

see and that makes
sense um agents that were less
optimistic were Expos opposed to more
negative violence events also something
we would expect to
see but what we can also see in this
plot is there is quite a lot of
variability uh but this is actually a
really good thing because when we're
thinking about um humans they have a lot
of variability so it's it's nice to have
a model that that can show this so
there's also instances um let's say an
example this so This agent here is
pretty optimistic but they were EXP to
lots of negatively valenced events so
this can be the things if someone well
the agents if they're initially exposed
to a lot of good things earlier on it
can kind of be protective so when they
get these negative Veen events later on
it doesn't change their model as much it
doesn't change their optimism as much so
we can think of that as
protective or we have agents here who
weren't exposed to many negatively
valenced events um but they don't have
much optimism and this could be because
maybe they're exposed to these events
much earlier on in the in the
simulation um and this yeah so this is
also with what what we would see we can
also see in in the literature is that
good things that happen can be
protective against the more negative
things that can happen so this with this
was called to say that they had the
variability so then this so that was the

optimism and now we're looking at the
likelihood matrices from the simulation
so this is the likelihood matricies that
look at the um optimism and pessimism
States versus the low and the high
arousal so here we have an agent who has
optimism level 07 so this is a pretty
pessimistic
agent and what we can see is that the
higher r
outcomes there's a lot there's a lot
higher Precision between P the
pessimistic State and the higher Ral
outcomes so what this means is that
these agents get higher risal outcomes
and it's like okay I'm in a pessimistic
state I'm in a pessimistic state so that
this this isn't this is um what we would
expect to see with someone who who is
anxious so this High arousal it's
getting giving a lot of information that
you're in in a certain
State and then here we have this
optimistic agent who has a completely
flat likelihood Matrix and this is a
healthy healthy U mapping because the
whether you're experiencing high or low
arousal doesn't um give you any you can
be in either State and that's fine um so
that's again how we would think of of
healthy
individuals uh they can experience High
arousal and that can be from a good or a
bad thing and they it doesn't give them
a signal or something's bad's happening
right
away so this was also we were excited to
see this um this kind of mapping in the

likelihood Matrix from the
simulation we also found a lot of
variability in these as well these are
just two examples because we ran 200
different
agents um but again variability for us
and what we're doing is a good thing so
this just gives us um yeah more ways we
can look at at different individuals so
that we were happy to see the
variability as
well so just to sum up experiment one um
we found that exposure to more negative
veence events reduced optimism which is
what we wanted to see we found that
pessimistic agents Le higher Precision
for higher arousal which inferred
pessimistic state and then we found
optimistic agents had a flat likelihood
Matrix so Rouser did not influence their
state so these were two things that that
we wanted to
find so now I'll go on to experiment
two so for experiment two we use the
belief updating task that's used in
literature so this is a standard
optimism task that's used used in in
humans and the way this task works is is
someone a participant is asked to
estimate the likelihood of a bad thing
happening so what's the likelihood um
you'll develop cancer and you give that
estimation and then the participant is
shown the actual likelihood of of that
estimation and that likelihood can be
lower or higher than their estimation so
if they're told their risk of of
something bad is lower that's considered

a good news trial because they've just
been given good news whereas if they're
told their risk of something is worse
that's a bad news trial because they've
just been given bad
news and you can look at how much
participants update to good news so how
much do they change their beliefs when
they get good news versus how much they
change their beliefs when they get bad
news so it's measur it's measured here
so we can see that this is the control
so controls are are optimistic because
people are are generally optimistic
and we can see that they update more to
good news than bad news and then we have
the um participants who have major
depressive disorder and they don't have
this same asymmetry when they when they
update to good and bad
news so we wanted to use this task uh to
see if the
optimism State factor that we simulated
or in this first experiment could then
be used used in the belief updating task
um to to show if we can see the results
that we find in the
literature because if we can then show
that uh well agents who are more
optimistic by by our model how we're
defining optimism update uh have this
same asymmetry and belief updating and
that pessimistic agents don't then our
model it's it's kind of more evidence to
show that that our model might be
capturing something that is that's
that's
useful so I'll go over the model here so

we set up the task so we had the
optimism and pessimism State Factor so
if a if the agent was in an optimistic
state it updated more to good to a good
outcome than a bad
outcome if the agent was in a
pessimistic state it updated more to a
bad outcome s a good outcome so this is
this the um likelihood Matrix
here the there was a initial hidden
State factor which was the belief about
the good versus bad
outcome so the agent gets gets this
belief about the good versus bad outcome
they're then fed observations which is
either um better than expected or worse
so good news or bad news so they give
good news or bad news
observations and then so then they are
updating their uh hidden State Factor
belief from the observation of good
versus bad
news and whether they're in an
optimistic or pessimistic State this
influences how much they update to good
versus bad
news so the optimism State factor is
conditioned on on this updating here
and then we took the difference between
the initial hidden State Factor belief
so that's here compared to the hidden
State Factor belief after the
observation so that's the D in then in
the next step and we took the difference
to look at the uh good versus bad news
for the
updating for this we actually ended up
running so usually in in the task that

human participants run they do 70 trials
so we wanted to do 7
trials the way that this best worked
though was to actually run 70 different
models so we so each model is um each
trial is a different
model and then we take the results of of
each of
those um which yeah I mean I guess that
can we can get into a discussion about
modeling and what we're modeling and how
it's working but I suppose you can think
of like well if we have a model model of
my belief about car accidents and we
have a model about my belief about
cancer they could all be different
models in for the for this purpose it's
not it's it's a different belief that
we're we're it's a different concept I
guess
um I hope this is all making sense it's
I haven't done an online talk for ages
and
it's and it's um hard to know how this
this is gone
Okay so we've got our belief updating
task and let's look at the
results so here we have our optimism
State Factor so this is the optimism
level so this is a very optimistic agent
this is a very not optimistic
agent and we find that the optimistic
agent updates more to good versus bad
news so we can see that here um we can
see that the more pessimistic agents
update more to bad versus good
news and then here we have this figure
from from the study that looked at

optimism in controls versus people with
major depression uh depressive
disorder and we have this here to show
that that we did kind of replicate this
this um pattern in in belief updating in
our simulation so that was exciting so
the major so if if these if the people
with MDD uh have more real depressive
realism than they would be this optimism
level5 and we can see yeah that we that
we are replicating this
result um so that was that was
exciting because this we were able to
show
that
so we found that our belief updating
task simulation results were comparable
to the literature and optimism
um so we say that this is kind of some
evidence for our optimism model because
we're able to replicate things that that
are already out
there and this also provides a good
basis for a model that can be tested for
fitting data so if you think about the
first when I was talking about the the
goals of of using models in the clinic
we want to have models that we can also
collect data and and fit them for and
not all models um you could are able to
do that so easily it can be it can be
tricky so this is a model that we can
then we can now just go and get people
to do this task and fit the model to see
if if we then get the same um optimism
State factor and we can also look at
their their likelihood matricies so
that's stuff that we can

do all right so now I'm on to the last
experiment so we have simulated the loss
of optimism or maintained it depending
on on the development of the agent we've
shown how it's um worked in this belief
updating task and now we're going to
look at optimistic action in this two
arm banded
task and this in this two Arm Bandit
task uh what we did was the agent could
either select left or
right um so left was highrisk high
reward right was low risk low reward or
there was a third option where the agent
could select stay so this was like the
safe option so if you selected stay you
didn't lose any money but you didn't win
any
money and this is an
important uh thing to be out of model
for what we're interested in
because if if if you are depressed and
you are not optimistic one of the things
that you can tend to do is is stay at
home and not go out and engage in all
those things that that I was talking
about at the start of the
talk so it's it's helpful if we can see
um to model this what what who would be
decide making this decision to stay stay
you know just stay safe stay at home not
but not getting any rewards
overall and one of the
uh this is kind of related to optimism
but one of the um interventions for
depression at the moment is behavioral
activation in cognitive behavioral
therapy which is the idea to get people

to go out in in the world more and
engage in more things so it it would be
good if we could have a task that we can
measure this this decision
making so that's what we that's what we
did so in this task yeah we have our
um our three actions that so so this
this model involved policies so uh we've
had a hierarchical model to begin with
with we had this second um our second
model had no policies and then this is
where the the agent this is where our
optimistic action comes
in and here again we have our optimism
versus pessimism State Factor so this is
what's been going on through out and so
here we have the the generative process
which is actually just the structure of
the two Arm Bandit task so if I select
left um we've got 70% chance of a large
win 30% chance of a large loss if I
select right we've got 60% chance of a
small win 40% chance of a small
loss so this is what's going on in the
actual task and this is the belief of
the agent so if you're optimistic you
have this High Precision over the large
wi and the small wind so just winds in
general and if you're pessimistic you
have this large Precision over the over
the loss over the losses rather than the
wins and this is controlled by this
State Factor here this is how the the
agents updating their
beliefs um and we ran I think it was 60
trials for for this for this
task so here are our results for this
task the total winnings and so again

we've got our optimism level so this
would be an agent with optimism level .9
optimism level .1 so this is pessimistic
agent
and we can see that the more optimistic
an agent is the more winnings they have
overall so in this task setting um
having this Precision over these
positive
outcomes was led to this these higher
winnings the the agent who had the
Precision over low outcomes actually
ended up losing
money but was also important about this
figure is that we can see that it wasn't
that the most optimistic agent had the
most winnings we see that the agent here
has has the most
winnings and a important idea when we're
thinking about optimism and
interventions is it probably isn't going
to be the case
that 100% optimism all the time in every
situation is a good thing different
environments that we're in there's going
to be different levels of optimism that
that are going to be most beneficial and
that's another thing that's what's so
nice about um being able to do these
models is that we can kind of explore
what those different levels look like
and um and and yeah test that out in
different environments rather than just
saying oh optimism is good let's make
everyone optimistic without actually
really understanding how it's going on
in this sense so findings like this I
are are

important so then we wanted to look at
well what are what's the actions of
these agents that led them to this
different
these different
outcomes so we have our pessimistic
agent here and we can see that they just
selected left and right and then they
just like no that's it I'm just going to
not I'm just going to stay so we can
think of this as somebody who might be
depressed who goes out has some bad
times and they're like no I'm just going
to stay at home now that's that's that's
not going to give me any any of Rewards
or anything that I'm looking
for uh we have this middle agent who
kind of went back and forth between
between the left and right arm and then
here we have the optimistic agent who
went right left and then just stayed in
the left arm which is the high risk High
reward but overall that would that would
lead you to the best um the mo the most
money in the
end uh so this this is nice because we
can see um what kind of actions uh the
actions of how of what kind of actions
would result from the different belief
belief that these agents have and this
is the behavior that we've been talking
about so our two Arm Bandit task uh was
able to capture optimistic
action and we could see that the
optimism promoted the engagement action
because the optimistic agents didn't
select stay they they kept going in the
task selecting left and

right we also found that full optimism
is not the most beneficial in this task
setting which is an important
consideration especially if we're
thinking about
interventions oh and then yep so they
were the three simulations that we
ran
so with this with these three different
simulations we feel like this is pretty
good uh initial support for an active
inference model of optimism
there's a lot of uh positives about
modeling optimism with this active
inference model we can use it in these
different task settings we can we've set
up models that we can now collect data
for um so that's the first
conclusion our model show demonstrated
how optimism could be lost in
development which is um an important
consideration uh belief updating task
grounded the model in literature and
optimism so we're able to show okay this
is how it fits to things that are
already out there measuring
optimism and our two unbanded task was
able to show how optimistic
action um was able to measure the
optimistic
action so the next step is to uh fit
participant data to the optimism model
and and we've designed these models with
that in mind um so a lot of the choices
that we made were for this next step uh
which is going to be an exciting step I
think to see if we can
um yeah get see if this this is we can

get data to fit and and then hopefully
use these models as a tool in the
clinic thank
you awesome
awesome um great well should I keep
sharing this um if somebody ask
something we can bring it back up but um
great thank you for the for the
presentation um yob do you want to give
any initial thoughts or
or no uh this is uh a result of Beth's
brain power and so I'll leave word to to
Beth on this
um okay
so a few questions I wrote down and then
also um we'll see what else people ask
so what what brought you from the review
of how people had modeled optimism to
this relationship with anxiety and
arousal so what so sorry so how why was
I interested in the optimism and the
arousal yeah okay
because it seems like you could be very
chill and have a a high believe that you
know the lottery is going to be better
than expected or something like that so
what what brought you into to looking at
the relationship with arousal
specifically yeah um that's a really
good question it was thinking about
events that happen in childhood and
development um that can C like so so
what happens when when a child has a
traumatic event what happens when
they're in an uncertain
environment um so initially that was the
idea and then we're thinking about this
arousal seems to play a part in in these

traumatic events and that is when um
when children are developing they are
learning this this kind of mapping so we
first think of that and then we looked
at the literature on optimism and
arousal and then it's like oh actually
optimism actually helps
maintain uh healthy arousal so it was
kind of it was just a search I guess and
finding thinking about what happens
during development that might affect
optimism what's the nature of those
experiences so what is it about those
experience that cause a child to feel a
certain way or have a certain belief
obviously there's something going on in
in the body those things colle uh
connected to anxiety and then
that yeah and then it happened to be
like oh this is this is cool and
interesting now because we're reading on
optimism and arousal and optimism is
good at maintaining that can we fit
those things together uh so we kind of
explored it in that way and I guess
what's also nice about running the
simulations
is when we come up with those
ideas uh when maybe not so they're
interesting ideas we're not 100% sold on
them and then you run the simulations
and then you're like oh okay cool this
we can see these Dynamics playing out um
which seems to tell us something
interesting yes definitely yeah
formalizing structurally different ways
of thinking about these psychological
constructs and and what about like any

situations where
pessimism or just sort of on the mark
accuracy like what is it about
situations or
Lifestyles or different areas where
optimism is relevant and it is it always
that way
or where wouldn't it be so if there's
situation where optimism is not a good
thing yeah yeah well I think it really
is important to think of an individual's
environment so I mean I live in
Melbourne I have a great life for me
it's like very a good thing to be
optimistic because if I go out into my
environment there's lots of
opportunities so it pays for me to think
things are better than they are because
I can go out and do all these things and
and there's lots of opportunities there
I wouldn't say someone in a a serious
like if someone's in a war zone I
wouldn't say oh you know what you should
be optimistic like that's an environment
where
optimism won't be beneficial and when
we're thinking about these interventions
that's such an important thing to
consider uh yeah so and also if we're if
so you could have a a patient who might
be in depressed or low optimism because
they're in something like a domestic
violence situation you wouldn't want to
say oh you should um just be optimistic
like that will help you want to help fit
their environment first so I think the
environment is such a big important part
in when optimism is a good thing um but

yakob and I have spoken about this a lot
if you so this is the idea of the
generative model and the generative
process if we are optimistic and we have
these these positive beliefs and then we
engage in these positive actions we do
also change we can influence the
generative process so we can we can
change our environment to be better as
well so it's like this it's kind of like
a a
feedback loop so there are situations
where being optimistic because your
environment affords opportunities but
then that optimism and engagement
changes the environment to afford even
more opportunities and make things
better
overall yeah make makes sense some
situations
[Music]
where the agent's beliefs or actions
don't change like the slot machine but
then there's other situations like maybe
interpersonally
where getting somebody else excited as
part of like a feedback loop that
that um okay I'll ask a question from
the chat upcycle Club
wrote do you think active inference can
help computational Psychiatry and even
cognitive science in general to model
additional cognitive biases beyond the
optimism
bias um that's a really good question
and yeah I think that
um we can model I think it lots of
different biases with active inference

and and other models as well and I think
that they're going to be really
important in um computational Psychiatry
because so much of our behavior and
um the things that we do of a lot of it
can be from the biases I mean we all
have them um all different biases and a
lot of them we're not aware of so a lot
of them we we think we don't have them
but we do and these models are a really
good way of of investigating that um I
have lots of biases I think I don't have
um and and yeah but if I could do a if I
could have a model of it and do a task
that could look into that without me
self-reporting on it I probably would be
very
helpful um and I guess the
the argument for using active
inference is when we are going out in so
the way that our the active inference
modeling works and we're doing actions
to kind of fulfill these beliefs that we
have it can kind of show how all these
different biases could be um sustained I
guess not just the optimism one others
as well because we're looking for
observations and and being in the world
to to to fit our
model yeah the piece about action as
like a self-fulfilling prophecy in
active inference is is interesting
because in a reward maximization model
of behavior selection it's almost like
optimism or at least do doing the best
of the options that are available that's
the nature of action selection itself
whereas an act of inference something is

being selected because it's believed to
be likelier or likeliest
so in that sense it sort of seems to on
a first pass encode this sort of ball
rolling downhill just down the middle
things should be just right as they are
so then it's interesting to see like
what are the ways that you you can model
optimism and then the fact that you had
it as a second
level sits it as a sort of domain
General feature or like cognitive
summary variable um so is this like the
same generative model that you adapted
three ways or or how were across the
experiments was it the same graph being
used in in slightly different
settings yeah it was the same the same
model so that was that and and you know
I again because we're trying to have a
model that has an application at the end
I mean ideally if we're thinking about
what this is really looking like it
would be heaps of levels of models and
optimism is that the at the very top but
but that's that wasn't our goal is our
goal was something that can be used but
that yeah I think optimism is definitely
this if we're thinking about
hierarchical models it would be kind of
very much at at the top and shaping our
um yeah perception and action on all
those things in in lots of different
ways um yob want to at add anything or
or what what brought you in the lab to
want to explore this
question other than having an optimistic
belief about a cool grad

student I mean it's it's just 100%
Beth's um initiative to focus on
optimism and
um the the group as such has a nice kind
of active inference Focus um on you know
trying to model various things and
contemp contemplative practices as well
and and which relates to mental health
and well-being and optimism is I've
discovered through Beth's work here
really a key element in this and then
there are some nice Beth and I have
talked a lot about how how might this
connect to the free energy principle and
so on and so we've kind of played rather
playfully around with ideas of a
universal optimism bias um that a
self-evidence and agent should have um
or might have so trying to so I don't
know if you mentioned Beth um that the
optimism bias is found in in an
astounding range of different creatures
um you know M donkeys you
knowes birds and so on and so it's not
like um you know status bias or things
that we we humans have it's not a
specifically human bias it's it's really
astoundingly Broad and so again Beth and
I have talked a lot about well that's a
kind of funny thing isn't it that it's
it seems more fundamental than other
biases or maybe other biases are kind of
linked to the optimism bias or something
like that and so
we've then found some some traces of
optimism bias uh in in the work from
Thomas par and and so on as well and
then we tried to kind

of rather speculatively draw a line
between the two so so it's a very good
example of this this lab this Center is
is very led by you know the the projects
that uh the PhD students and the postu
want to do and I kind of come for the
ride which is fantastic and then
something magical happens when when
these things begin to converge and so on
so it's not a coincidence of course that
that um Beth's work on optimism here
really kind of lifts things up and and
kind of gives us this kind of sense of
but there's something bigger going on as
well and then you know Beth's real Focus
as you started talking about Beth is how
can this work out there in the world in
the clinic and so on um which is also
incredibly important because it's very
easy to sit and just be you know super
enthusiastic about active influence and
the free energy principles what what
difference does it make um so that's
really nice as
well yeah the that's in interesting so
so what was the
universal optimism
bias you I talk about this with yeah
yeah
yeah so I think one one way to kind of
so this is written for a um a kind of
celebratory special issue of a journal
for Carl fiston 65th birthday some of
you might have seen and so it's a little
bit tongue and cheek a little bit
informal like these puff pieces
sometimes are I wouldn't call it puff
piece but um you know it's a little bit

of a a shangra and so on so we allow
ourselves to let our hair down a little
bit and and and just speculate a bit and
there's this nice um very short uh bit
in Thomas par and giovan psul and Carl
friston's active inference book
where and one of Thomas's earlier papers
as well where he says um you know
if the agent exists it must be able to
kind of withstand the dissipative forces
of the universe and then Thomas goes on
and says this means there's a kind of
univer a kind of optimism bias that must
be in play where in this coupled
illation between the agent and the
surrounding world that the agent kind of
believes that they are have the upper
hand and and we interpret that as a
universal optimism bi and then we try to
draw the line to this behavior that we
see in all these different kinds of of
creatures and organisms out there um to
put an idea out there more than to kind
of prove anything but um it was kind of
an interesting thing to think about
because then the optimism
bias we can kind of come around in a
circle between you know something very
fundamental about the fenity principle
and self- evidencing and so on and then
on the other end you know the focus of
Beth's work on on clinical relevance and
so on and maybe somehow arrive at
something
interesting that's cool it's kind of
like seeing the first principles
grounding for what what is seen in the
clinic and then if the beliefs

were if if if the uh be or the person or
whatever it is believes that things are
worse than they
are then that could be part of a
descending spiral so that the question
is how much optimism from sort of like
just a tiny bit to delusionally
risky and then it it it
um sets that as something about
differential persistence like you could
have whole populations and simulating um
Evolution and all this with different
kinds of um
biases um so what
where else slash next will you be
heading in the work
Beth um so we just have also put out a
preprint with some rat data some um it
was a study uh giving rats siloc cybin
which is the ingredient in magic
mushrooms and we did active inference
model fitting to that data to show that
over time um psilocybin was increasing
optimism in the
rats and uh for that we it's it's kind
of the same task Paradigm that the rats
can decide to engage in the task or not
so so similar to the the two unbanded
task I just spoke about um so that's
something I'm I'm super excited about
because obviously s Simon is getting a
lot of interest at the moment
with treatments for depression so I
think understanding the mechanisms of
that
is super important and so hopefully that
work can go to the next step when we're
looking to see how those models might

work in humans like the cocy and and
active inference models might work in
humans um then data fitting for
collecting data for the the studies I
spoke about
today um we also are coming up with a
computational model of how gratitude
practice might uh increase
optimism and again that that's all
thought out with how we can then collect
data and and fit and test that as well
so lots of different optimism
interventions and and
modeling how do you see epistemic value
in relationship to
[Music]
optimism um like learning and
updating um for its own sake rather than
expecting utility of a situation
well I think
that that is inherently
optimistic do you do you does that make
I just think that that's like that is an
optimistic thing to go out there and
reduce uncertainty and believe that
there's you know just for the sake of
like oh well there's going to be
something more I think that that that's
an optimistic action and belief in
itself
I think also what we've talked about
death often
is uh if you don't optimistically engage
you might be missing a state transition
out there in the environment that might
have happened without you knowing so you
keep going to parties all parties suck
and you keep you optimistically keep

going out there and part of it is well
maybe you know I go on the you know the
10th party there's been a bit of a
change in the social scene uh the toxic
person has left I have matured a bit now
it will be good so there's been these
State Transitions and if I don't engage
I won't learn like I won't get the
epistemic value about State transitions
up so that's a a kind of an element of
an epistemic side of optimism maybe as
well that's very interesting it's sort
of like it could be possible if and
especially what you brought in in the
third example with actions over
optimistic state so
basically attention or the ability to
control one's optimism State I mean
that'd be kind of cool but then there'd
be a way to
weave different foraging bouts so that
when you go out and it you go out with
high hopes and then when those High
Hopes are confirmed it's like great I
should be that optimistic but then when
it wasn't it could be framed as a
learning and then it it could just you
could stay in the game and continue
playing it's like I'm playing this slot
machine because I think it's going to
give a high expected reward and then I'm
going to play this one because I'm
trying to learn what its reward is and
then that might keep one doing that
engaged
action rather than that's very
interesting rather than only like
putting the thumb on the scale of um

utility and pragmatic value yeah exactly
um this is kind of related but we found
something really cool in the rats where
over time so it's like the last days of
the was like 14 days after the the
psilocybon
treatment the rats um who were given the
psilocybon had lower loss aversion so
that the in the C Matrix like the value
of of the loss was smaller for them so
the IDE is is that yeah you can go out
somewhere and even if you get something
bad it just doesn't affect you as much
you're kind of like you're more
resilient to that in a way and that's
the example of like y I can go to 10
parties and I have a bad time but if I'm
optimistic that that's not affecting me
and I and I'm yeah I'm still learning
I'm still going out there and I'm
curious to see like what the 11th party
is going to bring which is a good
thing it will it will be it it will be a
very interesting journey I mean you laid
out sort of the road map with going from
the the conceptualization of the problem
through the simulation and then that
will be interesting on the sort of
diagnostic sense making side and then on
the intervention side and then also so
many
existing um diagnostic instruments and
interventions
May kind of
fit into this way of um
modeling I I'll um ask a question from
the chat Courtney
asks would you consider a bias the same

as an expectation is optimism the
expectation of a positive
outcome yeah so that's a good question I
so like
it is an expectation of a good outcome
but then it also affects your it's not
just the expectation because it also
affects your belief
updating so it's not just oh I expect
the good things to happen when I get the
good things they're also changing my
beliefs more than um the bad things and
I think if it was only an expectation
you wouldn't have that asymmetry when
you're updating your belief so I think
that's why it's more just like that
that's probably more the bias component
to it it affects different things not
just it can be carried out in these
different ways not just that that one
way whereas if I just have positive
expectations that might not necessarily
mean when I get the positive thing I
update more to that positive outcome as
well but they're all good questions like
how these different things connect so I
would say positive expectations isn't
one element of the optimism bias
cool
um any sort of closing thoughts or what
what do you leave people with
Slash how could they learn more or or
build up their optimism bias or whatever
it may
be um well thank you so much for
inviting us on this has been so nice
it's been really cool it's so exciting
when you do your own work and you're

like how is this fitting to anything or
is anyone interested in this at all and
then you have nice conversations it's so
it's always really
positive um yeah I think it's uh I hope
that this kind of work gives people
hope and
I I I hope that there also there's
people who want to use active influence
in these in these more practical ways
it's been helpful and if anyone ever
wants to reach out and talk about ways
to do that I'm I'm happy CU I to do that
cuz I think active inference holds so
much promise as a framework and when we
can start um getting it out there in the
these kind of settings that's that's
going to be really cool um yeah and if
you can try and be
optimistic and if you can't we might
have some interventions for you
soon awesome yob any last
thoughts no I think it's appropriate
that Beth gets last uh word though
uh maybe wait until we publish the cying
paper before you take medic mushrooms to
become
mytic we'll do we'll
do okay thank you very much both um till
the next time and I'm sure it'll be a
good
one thanks so much Daniel see you bye
bye for